



OS4All — Digital Twin Architecture

Happiest Health Digital Twin Challenge 2026 • Master Technical Architecture Specification

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Use Case: Type 2 Diabetes / Near-Term Glucose Spike

1 DATA SOURCES & INGESTION

Multi-Stream

Stream 1: Static EHR

SYNTHEA FHIR

Longitudinal Patient Cohort:

- Demographics: Age, Gender, BMI
- SNOMED: Type 2 Diabetes (44054006)
- LOINC: HbA1c (4548-4), Fasting Glucose
- RxNorm: Active Metformin / Insulin therapies
- FhirImportService / FhirPatientMapper

Stream 2: Dynamic Telemetry

5-MIN SIMULATED

Continuous IoT Stream:

- CGM Interstitial Glucose (30–500 mg/dL)
- Glucose Velocity (dG/dt in mg/dL/min)
- Heart Rate (bpm) & RMSSD HRV (ms)
- Sleep Architecture & Cumulative Deficit
- Step Velocity & SpO2 Saturation
- DigitalTwinSimulationEngine (Scenario 1–5)

Stream 2b: Real Device

OPTIONAL

Android Health Connect:

- Read-only MethodChannel bridge (API 26+)
- Physical Steps, HR, HRV, Sleep, Glucose
- Zero writes; zero fabricated biometrics
- Graceful fallback to simulated on desktop/web

✓ Sensor boundary validation & null-safe packet ingestion

2 FUSION & BASELINES

Normalization

Dual-Stream Data Fusion

CORE INGESTION

DigitalTwinFusionEngine:

- Temporal alignment of EHR + 5-min IoT
- Sensor noise filtering & Tukey fences
- Unit normalization (mg/dL, bpm, ms, hrs)
- Zero-fabrication null imputation policy

Personalized Baseline

45-DAY ROLLING

"Normal for the individual, not population"

- Rolling 45-day evaluation window ($N \geq 5$)
- Bessel-corrected sample standard deviation (s)
- Normal operating tolerance envelope:

$$\text{Envelope} = [\bar{x} - 1.96 \cdot s, \bar{x} + 1.96 \cdot s]$$

Patient-Specific Z-Score Engine:

$$Z = (x_{\text{current}} - \bar{x}_{\text{baseline}}) / s$$

Glucose Z

HRV Delta %

RHR Drift

Sleep Debt

- PersonalBaselineService / Controller

✓ Continuous physiological envelope tracking

3 DETERMINISTIC TWIN ENGINE

Authoritative Source

Metabolic State Machine

4 STATES

• STABLE

• DRIFT (Sleep/Stress)

• ELEVATED RISK

• ACTIVE ANOMALY

Multi-signal perturbation transitions driven by baseline departures

Deterministic Risk Scoring

0–100 SCALE

- Glucose Velocity: 30%
- HRV Suppression: 20%
- Resting HR Drift: 10%
- Glycemic CV%: 20%
- Sleep Deficit: 15%
- Chronic BMI: 5%

2-Hour Glucose Trajectory

120-MIN PROJECTION

Damped Velocity Decay Formulation:

$$G(t) = G_0 + (v_0 + v_{\text{act}}) \cdot [(1 - e^{(-\lambda \cdot t)}) / \lambda]$$

- $\lambda = 0.010 \text{ min}^{-1}$ (T2D clearance) | v_{act} : GLUT4 walk modifier
- Milestones: $t = 0\text{m}, 30\text{m}, 60\text{m}, 90\text{m}, 120\text{m}$

Feature Contribution Table:

Exact points (+/-), direction (RISK / PROTECTIVE), physiological rationale

- MetabolicPredictionEngine (Authoritative Source of Truth)

🔒 Sole Authoritative Calculation Layer (100% Deterministic)

4 EXPLAINABILITY & AI

Grounded Layer

Google Gemini Layer

GEMINI-3.8-FLASH

Official Google GenAI SDK:

- SDK: com.google.genai.Client
- Input: Read-only structured twin telemetry
- Plain-language clinical synthesis
- Interactive Virtual Patient inquiry

- ✗ NEVER computes risk scores or biometrics
- ✗ NEVER invents medical diagnoses

Offline Fallback Engine

ZERO DOWNTIME

100% Offline Resilience:

- Active when API key unset or network offline
- Rule-based structured clinical narratives
- Zero hallucination guarantee
- Full hackathon judge reproducibility

Boundary: Gemini translates verified calculations; deterministic engine owns math.

- GeminiExplanationService.java

✓ Non-diagnostic grounded interpretation

5 USER EXPERIENCES (FLUTTER)

Cross-Platform

Patient Mobile UI

APPSHELL (5 TABS)

HOME TWIN LIVE AI PROFILE

Daily biometrics, 3D twin avatar status, continuous live curves, grounded AI counselor & 45-day baseline envelopes.

Doctor View — Clinical Workstation

DIRECT ACCESS

DoctorDashboardScreen (Reached via TWIN tab → "Doctor View")

- Cohort Selector:** Alex Rivera, Shara Senger, Marcus Vance
- Patient Hero Card:** Demographics, BMI, Twin State & Risk (0–100)
- 5 Interactive Simulation Scenarios:**
 - 1. Stable
 - 2. Sleep Drift
 - 3. High Activity
 - 4. Spike Risk
 - 5. Recovery
- Live Telemetry & 2-Hour Trajectory Chart:** 120-min forward curves with 0m, 30m, 60m, 90m, 120m milestone cards
- Feature Contribution Table:** Inspectable mathematical drivers
- Personalized Baseline Grid:** Side-by-side Z-score analysis
- Unified Event Timeline:** Longitudinal historical + live events
- Virtual Patient Q&A:** Clinician inquiry strictly grounded in telemetry

✓ Desktop, Web & Mobile responsive workstation

🔒 SUPPORTING ENTERPRISE INFRASTRUCTURE: SECURITY, PERSISTENCE & AUDIT TIER

Spring Boot 3.3.4 • Java 17/21 • Spring Security • Spring Data JPA • Flyway

Stateless Authentication

Bearer JWT filter chain, BCrypt password hashing (cost factor 10), role-based endpoint authorization.

Granular Consent Enforcer

Contract-level consent verification for DATA_STORAGE & AI_INFERENCE prior to data processing.

Tamper-Evident Audit Trail

Cryptographic SHA-256 chained hash logs linking prior records for all biometrics, OCR, and AI queries.

Persistence & API Gateway

MySQL 8.0 (Prod) / H2 (Test) with Flyway migrations. Springdoc OpenAPI 3.0 / Swagger documentation.